

Haseeb Ul Hassan

AI / Machine Learning Engineer

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Iqama: Transferable | SCE: Computer Science Specialist | Honors: Presidential Gold Medalist

Summary

AI/ML Engineer with 2 years of experience designing and deploying production AI systems across Computer Vision, NLP, and Generative AI. Built ForecastIQ, an end-to-end demand-forecasting pipeline saving \$200K+ annually, and delivered custom CV/OCR/LLM solutions for enterprise clients.

Skills

- **Programming Languages:** Python, C++, SQL
- **Core ML & Computer Vision:** PyTorch, TensorFlow, Scikit-Learn, Ultralytics YOLO, OpenCV, Pandas, NumPy, Matplotlib, OCR (Tesseract, EasyOCR, PaddleOCR, Google Cloud Vision)
- **Generative AI & LLMs:** OpenAI API, OpenAI Agents, Google Gemini, LangChain, RAG Pipelines, ChromaDB, Pinecone, Ollama, HuggingFace Transformers, Prompt Engineering
- **MLOps & Deployment:** FastAPI, Docker, Weights & Biases, MLflow, Git, Streamlit
- **Cloud, Databases & Tools:** AWS, GCP, PostgreSQL, SQL Server, Power BI, Linux, GitHub Copilot, n8n

Experience

VisionX - AI/ML Engineer

Nov 2024 - Present

New York, USA (Remote)

- Designed, built, and deployed ForecastIQ end-to-end: a demand forecasting and automated planning pipeline that replaced manual spreadsheet workflows, saving \$200K+ annually in labor and reducing excess inventory.
- Improved a production computer vision detection model through targeted data augmentation and hyperparameter fine-tuning, shipping a version faster and more accurate than the deployed baseline.
- Delivered an automated image-based cataloging and data-extraction system for a major US enterprise client, combining a custom-trained YOLO OBB model, OCR, and LLM-based classification to reduce manual entry and improve data accuracy.
- Lead ongoing ML initiatives across forecasting, automation, and deployment workstreams, mentor a team of ML interns through daily standups and weekly demos.
- Integrated generative AI workflows (LLM prompt engineering, OpenAI and Google Gemini SDKs) into internal tools and client deliverables.

VisionX - Machine Learning Intern

Aug 2024 - Oct 2024

New York, USA (Remote)

- Trained a PaddleOCR detection model on a custom-labeled dataset, improving text detection performance.
- Fine-tuned LayoutLMv3 for document sequence classification, tuning batch size and learning rate for accuracy gains.
- Trained and benchmarked detection vs. segmentation models, converted label formats and applied augmentation to improve generalization.
- Developed Android applications integrating computer vision models for on-device detection and segmentation.

Projects

Automated Book Spine Detection & Metadata Extraction

- Built an end-to-end CV + OCR + NLP/LLM pipeline: custom YOLO OBB model for rotated book spine detection, multi-backend OCR (Tesseract, EasyOCR, Google Cloud Vision), and a custom NER model for structured field classification, refined via LLM post-processing and prompt engineering.
- Deployed as a Streamlit application on Streamlit Community Cloud, engineered pipeline caching to eliminate redundant detection runs on user interactions.

ForecastIQ - Demand Planning & Forecasting Pipeline

- Built a flexible-horizon (24 to 54 week) SKU-level forecasting pipeline with a Python planning engine that applied business rules from structured notes files to derive purchasing formulas for the client procurement team.
- Deployed on a client-provisioned VM accessed over VPN, served via FastAPI with automated daily execution.

Education

Institute of Space Technology (IST) - B.S. Computer Science

Sep 2020 - Jul 2024

Islamabad, Pakistan

CGPA: 3.77/4.0 - Presidential Gold Medalist

Certifications

IBM - Deep Learning with TensorFlow & Keras | Supervised ML: Regression and Classification | DeepLearning.AI - AI For Everyone | Python for Data Science, AI & Development | Intro to DL & Neural Networks with Keras